Unit 5
Storage Devices 7 hours

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Storage Devices

…. (Refer earlier chapters for definition)

Memory

Computer memory is the storage space in computer where data is to be processed and instructions required for processing are stored. It is used to store data and instructions.

The memory is divided into large number of small parts called cells. Each location or cell has a unique address which varies from zero to memory size minus one. For example, if computer has 64k words, then thismemory unit has 64 * 1024=65536 memory locations. The address of these locations varies from 0 to 65535.

Memory is primarily of three types

- Cache Memory
- Primary Memory/Main Memory
- Secondary Memory

Cache Memory

Cache memory is a very high-speed semiconductor memory which can speed up CPU. It acts as a buffer between the CPU and main memory. It is used to hold those parts of data and program which are most frequently used by CPU. The parts of data and programs are transferred from disk to cache memory by

operating system, from where CPU can access them.

Advantages

The advantages of cache memory are as follows:

- Cache memory is faster than main memory.
- It consumes less access time as compared to main memory.
- It stores the program that can be executed within a short period of time.
- It stores data for temporary use.

Disadvantages

The disadvantages of cache memory are as follows:

- Cache memory has limited capacity.
- It is very expensive.

Primary Memory (Main Memory)

Primary memory is an essential memory, which is accessed directly by the CPU for processing. It is a working space of CPU from where the CPU uses data for calculation, also stores intermediate or final results unless stored in secondary memory

Characteristics of Main Memory

- These are semiconductor memories
- It is known as main memory.
- Usually volatile memory.
- Data is lost in case power is switched off.
- It is working memory of the computer.
- Faster than secondary memories.
- A computer cannot run without primary memory.

Secondary Memory

The Secondary Storage also known as Auxiliary Storage, is used to take care of the limitations of the primary storage. That is, it is used to supplement the limited storage capacity and the volatile characteristic of primary storage. This is because secondary storage is much cheaper than primary storage and it can retain information even when the computer system is switched off. The secondary storage is normally used to hold the program instructions, data and information of those jobs on which the computer system is not working on currently, but needs to hold them for processing later. Themost commonly used storage medium is the magnetic disk. This type of memory is also known as external memory or non-volatile. It is slower than main memory. These are used for storing

data/Information permanently. CPU directly does not access these memories instead they are accessed via input-output routines. Contents of secondary memories arefirst transferred to main memory, and then CPU can access it. For example: disk, CD-ROM, DVD etc.

Characteristic of Secondary Memory

- These are magnetic and optical memories
- It is known as backup memory.
- It is non-volatile memory.
- Data is permanently stored even if power is switched off.
- It is used for storage of data in a computer.
- Computer may run without secondary memory.
- Slower than primary memories.

Types of Primary Memory

Primary Memory is of two types from the technological point of view and these are analog memory anddigital memory. Each of Primary memory is discussed below.

- **Analog Memory**

  These types of memories were used in the computer built during 1960's. The basic storage device in these types of memories consists of a small ring-shaped piece of magnetic material called magnetic core. Each magnetic core in a memory is capable of storing one binary digit of digital information.

- *Digital Memory*

  Semi-conductor or digital memories are static, lighter, and cheaper than magnetic core memories. These are volatile and data stored in them is lost when power supply to the machine is switched off. Digital memories can be divided into the following two types and these are described in the following subsections:

- *Random Access Memory (RAM)*

  RAM (Random Access Memory) is the internal memory of the CPU for storing data, program and program result. It is read/write memory which stores data until the machine is working. As soon as themachine is switched off, data is erased.

  Access time in RAM is independent of the address that is, each storage location inside the memory is as easy to reach as other locations and takes the same amount of time. Data in the RAM can be accessed randomly but it is very expensive.

RAM is volatile, i.e. data stored in it is lost when we switch off the computer or if there is a power failure. Hence a backup uninterruptible power system (UPS) is often used with computers. RAM is small, both in terms of its physical size and in the amount of data it can hold.

Different types of RAM are:

-Static RAM (SRAM)

The word static indicates that the memory retains its contents as long as power is being supplied. However, data is lost when the power gets down due to volatile nature. SRAM chips use a matrix of 6-transistors and no capacitors. Transistors do not require power to prevent leakage, so SRAM need not have to be refreshed on a regular basis.

Because of the extra space in the matrix, SRAM uses more chips than DRAM for the same amount of storage space, thus making the manufacturing costs higher. So SRAM is used as cache memory and has very fast access.

Characteristic of the Static RAM

-It has long life

-There is no need to refresh

-Faster

-Used as cache memory

-Large size

-Expensive

-High power consumption

-Dynamic RAM (DRAM)

DRAM, unlike SRAM, must be continually refreshed in order to maintain the data. This is done by placing the memory on a refresh circuit that rewrites the data several hundred timesper second. DRAM is used for most system memory because it is cheap and small. All DRAMs are made up of memory cells which are composed of one capacitor and one transistor.

Characteristics of the Dynamic RAM

-It has short data lifetime

-Need to be refreshed continuously

-Slower as compared to SRAM

-Used as RAM

-Lesser in size

-Less expensive

-Less power consumption

ROM (Read Only Memory)

ROM stands for Read Only Memory and it is a permanent type memory. Its contents are not lost whenpower supply of computer is switched off. The user cannot write into a ROM. Its contents are written at manufacturing time. ROMs store permanent programs and other types of information which are needed by the computer to execute user programs. ROMs store function such as sine, cosine, logarithm, square root, exponential and code conversion tables, etc. An example of a ROM is Toshiba Mask ROM, TCS 534000, 512 KX8 bits again.

Different types of ROM are:

- *MROM (Masked ROM):*

The very first ROMs were hard-wired devices that contained a pre-programmed set of data or instructions. These kinds of ROMs are known as masked ROMs which are inexpensive.

- PROM (PROGRAMMABLE READ ONLY MEMORY)

PROM is read-only memory that can be modified only once by a user. The user buys a blankPROM and enters the desired contents using a PROM program. Inside the PROM chip there are small fuses which are burnt open during programming. It can be programmed only once and is not erasable

- EPROM (ERASABLE PROGRAMMABLE READ ONLY MEMORY)

The EPROM can be erased by exposing it to **ultra-violet light for a duration of up to 40 minutes**. Usually, an EPROM eraser achieves this function. During programming, an electrical charge is trapped in an insulated gate region. The charge is retained for more than ten years because thecharge has no leakage path. For erasing this charge, ultra-violet light is passed through a quartz crystal window (lid). This exposure to ultra-violet light dissipates the charge. During normal use thequartz lid is sealed with a sticker.

- EEPROM (ELECTRONICALLY ERASABLE PROGRAMMABLE ROM)

The EEPROM is programmed and erased electrically. It can be erased and reprogrammed about ten thousand times. Both erasing and programming take about 4 to 10 ms (milli second). In EEPROM, any location can be selectively erased and programmed. EEPROMs can be erased one byte at a time, rather than erasing the entire chip. Hence, the process of re-programming is flexible but slow.

Types of Secondary Memory

Floppy Disk

A floppy disk is made of flexible plastic that is coated with iron oxide. The flexible disk is placed within a square cardboard jacket, or the jacket is often referred as cartridge. The cardboard jacketgives protection to the outer disk surface. Moreover, it has a liner which gives a wiping action to remove dust particles from the disk surface because it is harmful to disk surface as well as to the read/write head. The disk is loaded in the disk drive along with its jacket. Information is read or written through a small hole in the jacket. The jacket can be easily loaded into and unloaded from a disk drive whenever necessary. The read/write head of a floppy disk make direct contact with the disk surface during processing and that's why floppy disk gets worn because of its constant use. Floppy disks are generally found in 3, 5.25, 8 inches in diameter. Floppy disks can be coated with iron oxide either one side or both side of the disk. One can record anythingon each side of the disk. Floppy disks are cheaper than other types of storage devices. The capacity of disk surface of floppy disk varies depending on their diameter and mode of data transfer. The capacity of a single-sided-single-density 5.25-inch disk surface may only be a little over 100 kilobytes, while the capacity of double-sided-double-density 8-inch disk surface may be almost 2 megabytes.

Magnetic Disk or Hard Disk

This storage device has very much higher storage capacity than floppy disk or other types of storage devices. The principle of operation of hard disk or magnetic disk is similar to that of a gramophone recorder. A magnetic disk is a circular disk coated with magnetic oxide for easy magnetization. Alarge number of disks are stacked firmly on a common spindle. A motor rotates the spindle at a rapid rate. To access each surface of every disk a read/write head is there that is attached to a common arm. The read/write head can move in between inner most track to outer most track along with the arm.The heads do not touch the disk but float on air cushion at a few microns apart. Arrangement of datain each surface of hard disk can be divided in several tracks and each track can be further divided into several sections called sectors. Each sector has fixed size but the number of sectors in each track may vary. Each sector has specific address for data stored in these sectors. A track in a given sector nearthe circumference is longer than a track near the center of the disk. If bits are recorded with equal density, some tracks will contain more recorded bit than other.

In order to access information from a disk, the disk address of the desired data has to be specified. Thedisk address is specified in terms of the track number, the surface number, and the sector number. Information is always written from the beginning of a sector and can be read only from the track beginning. As soon as a read/write command is received by the disk unit, the read/write heads are firstpositioned on to the specified

track number by moving the arm assembly in the proper direction. Oncethe heads are positioned on the desired track, the head on the specified surface is activated, and hence it transfers the data to the primary memory.

Magnetic Tape

A magnetic tape transport consists of the electrical, mechanical, and electronic components to providethe parts and control mechanism for a magnetic-tape unit. The tape itself is a strip of plastic coated with a magnetic recording medium. Bits are recorded as magnetic spots on the tape along several tracks. Usually, seven or nine bits are recorded simultaneously to form a character together with a parity bit. Read/write heads are mounted one in each track so that data can be recorded and read as a sequence of characters.

Magnetic tape unit can be stopped, stated, to move forward or in reverse, or can be rewound.However, they cannot be started or stopped fast enough between individual characters. For this reason, information is recorded in blocks referred to as records. Gaps of unrecorded tape are inserted between records where the tape can be stopped. The tape starts moving while in a gap and attains its constant speed by the time it reaches the next record. Each record on tape has an identification bit pattern at the beginning and end. By reading the bit pattern at the end of the record, the control recognizes the beginning of the gap. A tape unit is addressed by specifying the record number and the number of characters in the record. Records may be of fixed or variable length.

Optical Disk (CD-ROM)

A CD-ROM is called compact disk read only memory. It consists of a rotating disk which is coated with a thin metal that is highly reflective. An intense laser light is focused on reflective surface of the rotating disk. Data recording is accomplished by turning the laser beam. The laser beam is turned on and off at a varying rate, because of tiny holes called PITS is burnt into the metal coating at the disk along its tracks. A tiny PIT represents 1 and a surface without a PIT is known as land represented by 0.In order to record to read the recorded data a less powerful laser beam is focused on the disk surface. In case of write enable CD-ROM separate laser beam are implied. The storage density of CD-ROM is enormous, the storage cost is extremely low, and the access time is relatively fast. It is used as good economical backup storage for any short of data and information. Since optical disk system can store as much as much as hundred times the contents of current magnetic tape drives. One small inexpensive disk will be able to replace 25 to 30 reels of magnetic tape and any data on the disk canbe accessed within a few milliseconds.

Punched paper tape

This type of storage device was used during 1980’s. It was a sequential-access secondary memory. In this

type of storage device data is coded on paper tape in the form of punched hole combinations. It was 1 inch wide and its length was of any size even 100th feet long. It was found in the form of rolls. There was a line of sprocket holes in the middle of the tape. The use of sprocket hole was to feed the tape through the tape punching and reading device. Information is recorded as holes punched in rows across the width of the tape. The one row of a punched paper tape represented a character. The maximum number of holes per row was referred to as the number of channels on the tape. An eight-channel tape was commonly used. The number of characters that can be stored in a given physical space was low for paper storage. Moreover, the paper tape media is easily torn and mutilated that is why the use of paper media as a secondary storage device was vanished day by day.

Magnetic drum

Like the magnetic disk, there is another storage device which is called magnetic drum is a directaccess storage device that can be used for both sequential and random processing. It is made of a cylinder whose outer surface is coated magnetic material. There is a motor which rotates the cylinder on its own axis of rotation at a constant and high-speed. The surface of the cylinder is divided in several tracks. Data is stored on those tracks as magnetic spots similarly as in the case of a magnetic disk. The recorded data on the tracks is read through a set of stationary read/write heads which are placed little distance apart from the surface of the drum. Whenever new data is recorded on the surface of the drum previously stored data gets erased. The data recorded on the surface of the drum will remain indefinitely until they are erased. The drum can rotate to a speed of few hundred toseveral thousand rotations per minute to produce a fast access time. The size and storage capacity of magnetic drums are very high. A small size drum having 20 to 30 tracks can store 5000 bytes. A drumhaving 500 to 1000 tracks can store up to 100 megabytes. Larger size drums rotate about its axis slower than that of smaller size drums, because of that the access time of larger drums decreases asthe drum speed increases. Magnetic drums have fast data transfer but are more limited in capacitythan magnetic disk that is why they are not used nowadays.

Mass storage

Mass storage systems are the storage systems that provide access to the hundreds of billions of bytesof stored data. They combine the advantage of both tape and disk technology. The storage medium is essentially a length of flexible plastic material upon which short strips of magnetic tape are mounted. These strips are then placed in cartridges, and the cartridges are loaded into a storage device that is online to the CPU. The same read/write technique is used as used with magnetic tape.

Flash Memory

Flash Memory (sometimes called “flash RAM”) is a type of constantly – powered non-volatile memory that can be erased and reprogrammed in units of memory called blocks. It is a variation of electrically erasable programmable read only memory (EEPROM) which, unlike flash memory, is erased and rewritten at the byte level, which is slower than flash memory up dating. Flash memory is often used to hold control code such as the basic input/output system (BIOS) in a personal computer. **When** BIOS needs to be changed (rewritten), the flash memory can be written to in block (ratherthan byte) sizes making it easy to update on the other hand, flash memory is not useful as random access memory (RAM) because RAM needs to be addressable at the byte (not the block) level.

Flash memory gets its name because the microchip is organized so that a section of memory cells is erased in a single action or flash. Flash memory is used in digital cellular phones, digital cameras, LAN switch, PC cards for notebook computers, digital set up boxes, embedded controllers and other devices.

Difference between Primary Memory and Secondary Memory

S.N.	PRIMARY MEMORY	SECONDARY MEMORY
1	Primary memory can directly communicate with CPU (Central Processing Unit) hence it is called main memory.	Secondary memory cannot directly communicate with CPU hence it is called auxiliary memory.
2	Primary memory is relatively faster than secondary memory.	Secondary memory is comparatively slower than primary memory.
3	Primary memory is needed basically by the system itself for processing functions.	Secondary memory is needed basically by users for storing of data and information permanently.
4	Primary memory is relatively much costly than secondary memory.	Secondary memory is comparatively muchcheaper than primary memory.
5	Primary memory has less memory space as compared to the secondary memory.	Secondary memory has a huge memory space as compared to primary memory.